

Machine Learning-Based Late Delivery Risk Prediction in Global Supply Chain Operations

Project Overview

This project presents an end-to-end Machine Learning and Business Intelligence solution for predicting late delivery risks in global supply chain operations. The solution combines data preprocessing, exploratory data analysis, predictive modeling, and an interactive Streamlit dashboard to support supply chain analytics and delivery risk assessment.

Project Highlights

- Developed a late delivery risk prediction system using Machine Learning techniques.
- Analyzed more than 180,000 supply chain records across multiple business dimensions.
- Implemented an interactive multi-page Streamlit dashboard for analytics and prediction.
- Identified high-risk delivery orders and key delivery risk factors.
- Provided business intelligence insights and operational recommendations.

Model Performance

- Best Performing Model: Random Forest
- Accuracy: 79.19%
- ROC-AUC Score: 88.93%
- Prediction Type: Binary Classification
- Machine Learning Features Used: 29

Key Risk Insights

- Total High-Risk Orders Identified: 11,390
- High-Risk Order Percentage: 31.55%
- Top Risk Driver: Days for Shipment (Scheduled)

Business Outcomes

- Early identification of potential late deliveries.
- Improved visibility into delivery risks across supply chain operations.
- Support for proactive operational planning through predictive analytics.
- Interactive KPI monitoring and executive-level business intelligence reporting.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn

- Streamlit
- Plotly
- Joblib
- Git & GitHub

Final Conclusion

Random Forest achieved the highest predictive performance among the evaluated models and was selected as the final model for deployment. The proposed solution enables late delivery risk prediction and provides actionable supply chain insights to support data-driven operational decision-making.